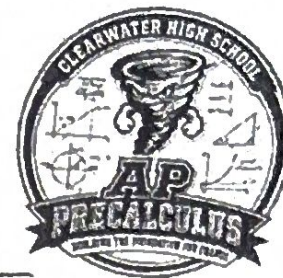


Lesson 1.7: Rational Functions and End Behavior



Topic 1.7 Rational Functions and End Behavior

Rational Functions

Let $N(x)$ (numerator) and $D(x)$ (denominator) be polynomial functions, where $D(x) \neq 0$.

Then, a Rational Function is: $R(x) = \frac{N(x)}{D(x)}$

The domain is all real # except where $D(x) = 0$



End Behavior for Rational Functions

The end behavior of a rational function can be found by the leading terms of the numerator and the denominator.

$$R(x) = \frac{ax^n}{bx^d}$$

Case:

Example:

Case I: degree of numerator < degree of denominator ($n < d$)

The line $y = 0$ is the horizontal asymptote.

$$R(x) = \frac{2x^0}{x^1 - 4}$$

$$y = 0$$

Case II: degree of numerator = degree of denominator ($n = d$)

The ratio of the leading coefficients is the horizontal asymptote.

That is, $y = \frac{a}{b}$

$$R(x) = \frac{6x^2 - 3x}{2x^2 + x - 5}$$

$$y = \frac{6}{2}$$

$$y = 3$$

Case III: degree of numerator > degree of denominator ($n > d$)

The graph has the end behavior $y = \frac{a}{b}x^{n-d}$

$$R(x) = \frac{4x^4 + 3x^3 - 2x}{2x^3 - 8x^2 + 3}$$

$$y = \frac{4}{2}x^{4-3}$$

$$y = 2x$$

Note: If the degree of the numerator is exactly one more than the degree of the denominator then the rational function will have a slant asymptote.

EX #1: Determine if the rational function has a horizontal asymptote, slant asymptote, or neither.
Provide a reason. If the function has a horizontal asymptote, write the equation.

A. $f(x) = \frac{1}{2x^2 + 3x - 15}$

HA: $y = 0$
 $x^0 < x^2$

B. $f(x) = \frac{3x - 3}{x + 3}$

HA: $y = 3$
 $x^1 = x^1$

C. $f(x) = \frac{3x^2 + 3x - 18}{3x - 5}$

SA
 $x^2 > x^1$

EX #2: Write limit statements to describe the end behavior for each function.

A. $f(x) = \frac{4x^3 + 5x - 3}{6x^3 - 9}$ $\frac{4}{6} = \frac{2}{3} = y$

Left: $\lim_{x \rightarrow -\infty} f(x) = \frac{2}{3}$

Right: $\lim_{x \rightarrow \infty} f(x) = \frac{2}{3}$

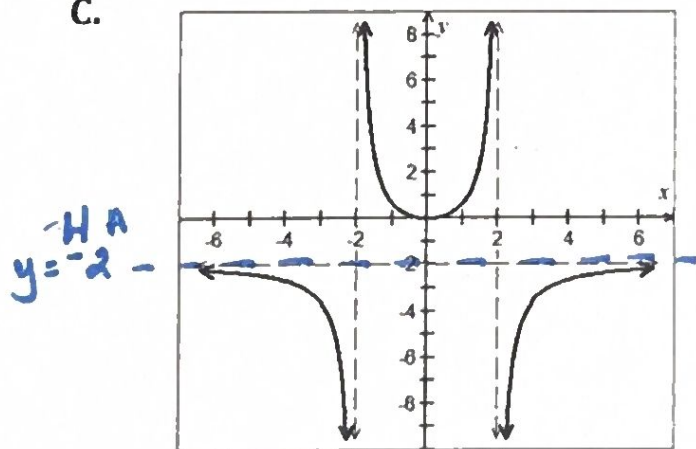
B. $g(x) = \frac{2x^4 - x^2 + 3x - 1}{x^3 - 8}$

$y = 2x$

Left: $\lim_{x \rightarrow -\infty} g(x) = -\infty$

Right: $\lim_{x \rightarrow \infty} g(x) = \infty$

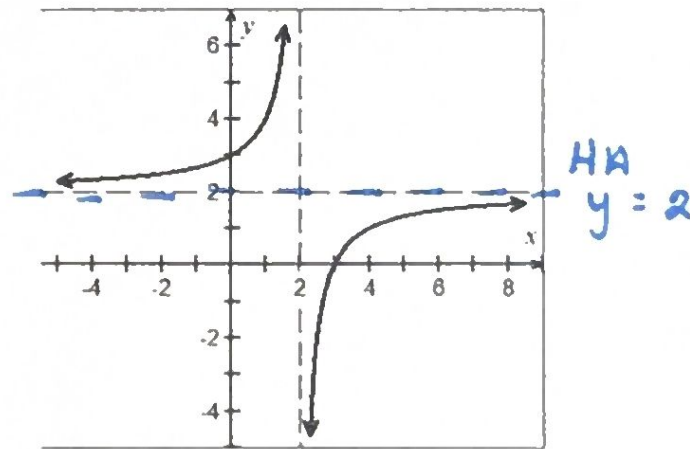
C.



Left: $\lim_{x \rightarrow -\infty} y = -2$

Right: $\lim_{x \rightarrow \infty} y = -2$

D.



Left: $\lim_{x \rightarrow -\infty} y = 2$

Right: $\lim_{x \rightarrow \infty} y = 2$

Slant Asymptotes

If the degree of the numerator is exactly one greater than the degree of the denominator, a rational function will have a slant asymptote that is parallel to the ratio of leading terms.

$$R(x) = \frac{ax^n + \dots + c_n}{bx^d + \dots + c_d}$$

Then, $R(x)$ will have a slant asymptote parallel to the line $y = \frac{a}{b}x$

When the numerator is divided by the denominator, the equation of the slant asymptote is $y =$ the quotient, without the remainder,

EX #3: Find the slant asymptote for the given functions.

A. $f(x) = \frac{2x^3 + 3x^2 - 4x + 1}{x^2 - 2x + 2}$

$y = 2x + 7$

$$\begin{array}{r} \overline{2x + 7} \\ x^2 - 2x + 2 \overline{) 2x^3 + 3x^2 - 4x + 1} \\ \underline{-2x^3 + 4x^2} \\ 7x^2 - 4x \\ \underline{-7x^2 + 14x} \\ 10x + 7 \end{array}$$

* ignore remainder when solving for SA

B. $g(x) = \frac{3x^2 + 5x - 1}{x + 2}$

$x + 2 = 0$
 $x = -2$

$$\begin{array}{r} -2 \overline{) 3 \quad 5 \quad -1} \\ \underline{-6 \quad 2} \\ 3 \quad -1 \quad | \quad 1 \end{array}$$

remainder

$y = 3x - 1$

* can only use synthetic division when the divisor is $x \pm c$

EX #4: Describe the end behavior for the graph of the rational functions. Analyze using limit notation to explain your answer.

A. $f(x) = \frac{2x^5 - 4x}{x^3 + 2}$

Consider leading terms only

$$\frac{2x^5}{x^3} = 2x^2$$

$$\lim_{x \rightarrow -\infty} (2x^2) = \infty$$

$$\lim_{x \rightarrow \infty} (2x^2) = \infty$$

B. $g(x) = \frac{3x - 5x^4}{x + 2}$

$$\frac{-5x^4}{x} = -5x^3$$

$$\lim_{x \rightarrow -\infty} (-5x^3) = \infty$$

$$\lim_{x \rightarrow \infty} (-5x^3) = -\infty$$